

(LMC)

AFP v19

2026-07-20

1

1.1

Pythia-1.4B “ ”
“ ” “ A ” “ B ”

- “ ”
-

2

2.1 (Model Parameters / Weights)

“ ”Pythia-1.4B 13.1
“ ” “ ” /
 θ θ theta θ_0

2.2 (Fine-Tuning / FT)

Pythia-1.4B → →
Full-FT “ — ” 13.1

2.3 (Weight Displacement / ΔW)

$$\Delta W(\theta, \theta_0) = \frac{\|\theta - \theta_0\|_2}{\|\theta_0\|_2} \times 100\%$$

- $\|\theta - \theta_0\|_2$ “ ” ℓ_2 —
- $\|\theta_0\|_2$
- 100%

$$\Delta W = 1.4\% \qquad 1.4\%$$

2.4 (Linear Mode Connectivity / LMC)

A B

- $\alpha = 0$ 100% A + 0% B →
- $\alpha = 0.5$ 50% A + 50% B →
- $\alpha = 1$ 0% A + 100% B →

$\alpha = 0$ $\alpha = 1$ A B “ ”

= =
→ LMC
→ LMC

2.5 LMC (LMC Barrier / $\mathcal{B}$)

.. ..

$$\mathcal{B}(\theta_A, \theta_B) = \max_{\alpha \in [0,1]} \mathcal{L}(\theta(\alpha)) - \frac{\mathcal{L}(\theta_A) + \mathcal{L}(\theta_B)}{2}$$

- $\alpha = 0 \quad \alpha = 1 \quad \text{loss}$
-
-
- bar
- bar ≈ 0
- bar ≈ 0.05
- bar ≈ 0.2
- bar $\approx 1.0+$

2.6 (Loss / $\mathcal{L}$)

.. ..

BCE (Binary Cross-Entropy) .. / .. / BCE

2.7 vs

(Within-domain barrier)

$\rightarrow$

(Cross-domain barrier)

$+$ $\rightarrow$

2.8 Per-Block

Transformer Pythia-1.4B 24 Transformer layer 0 layer 23

r Pearson

- $r = 1.0$
- $r = 0.0$
- $r = 0.995$ Pythia

2.9 Gaussian

Gaussian

2.10 (Layer-Selective Interpolation)

-
- Early (0–7) 8
 - Mid (8–15) 8
 - Late (16–23) 8
- 75% 8 8

2.11

U (Inverted-U) _____

-
-
-

(Monotonic) _____

-

2.12 Hessian

Hessian $\mathbf{H}$ " "

(Loss Landscape) " "

- (Wide Basin)
- (Narrow Basin)

2.13 SGD Peclet

SGD () " "

Peclet (Pe) " " " "

- Pe $\rightarrow$
- Pe $\rightarrow$

/

3

3.1

Table 1:	
Pythia-1.4B 13.1	
(Full-FT)	
~55K	+ ~55K
/	
bfloat16	
AdamW	SGD
NVIDIA DGX Spark 121GB	

3.2

1.

(Standard)

$\Delta W \approx 1.4\%$
2.

(High)

$\Delta W \approx 8\%$
- " " " "

—

6

2-5

3.3

- General
- Math
-

4

4.1

					3
					0.048
					0.053
					0.147
LMC	•	•	•	•	

4.2

				(Code)	0.048
				(General)	0.071
				(Math)	0.087
				(Medical)	0.147
•	•				

4.3 U vs

0.029 → 0.043 → 0.033 " U "

0.173 0.218

U
•
•